



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Pathology: Dilated and Hypertrophic Cardiomyopathies

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Session Objectives

1. Describe, and illustrate the **general** characteristics of **Dilated Cardiomyopathy** as well as its **gross** and **histological** features.
2. Explain the varied **etiologies**, pathogenesis and pathophysiology, along with **clinical** presentation, diagnosis, management, prognosis, and genetic characteristics of **Dilated Cardiomyopathy**.
3. Discuss the general characteristics along with clinical manifestations, genetic characteristics, and pathophysiology of **Arrhythmogenic Right Ventricular Cardiomyopathy**.
4. Describe, and illustrate the **general** characteristics of **Hypertrophic Cardiomyopathy** as well as its **gross** and **histological** features.
5. Explain the varied **etiologies**, pathogenesis and pathophysiology, along with **clinical** presentation, diagnosis, management, prognosis, and genetic characteristics of **Hypertrophic Cardiomyopathy**.
6. **Compare** Dilated to Hypertrophic Cardiomyopathy

Cardiomyopathies are heart muscle diseases

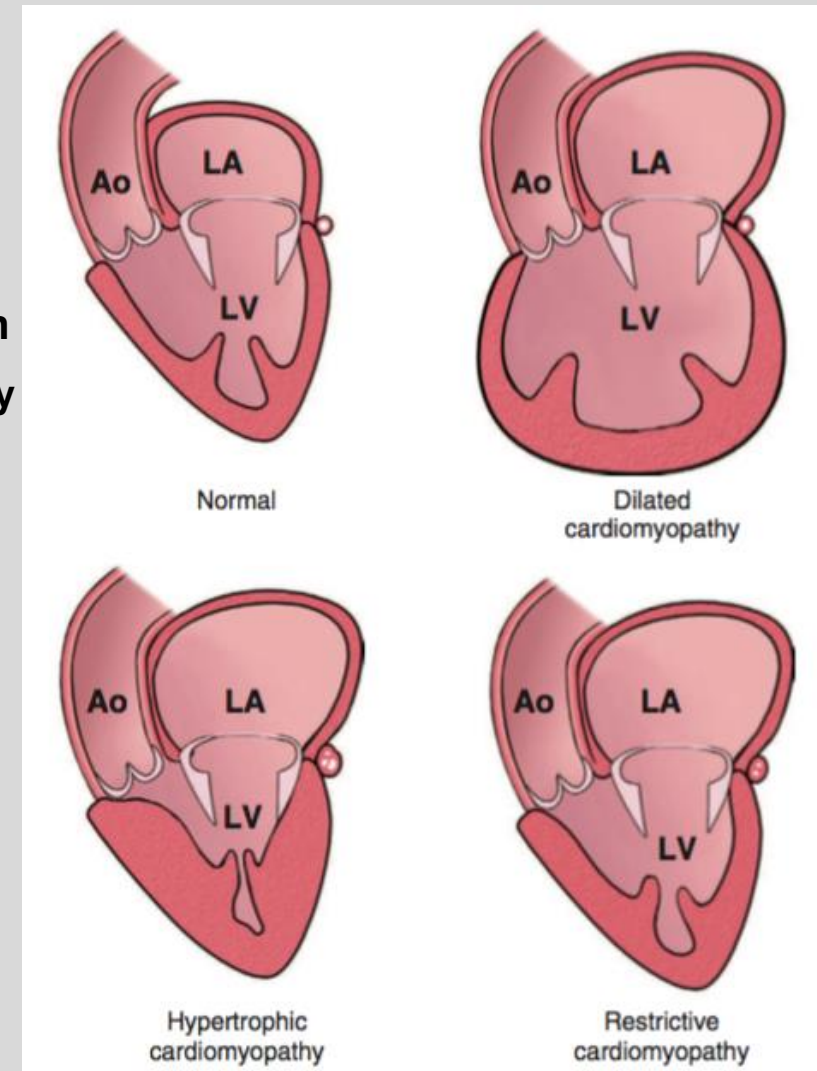
- **Generally structural** cardiac abnormalities manifesting as failure of myocardial performance
 - **Distinct from** impairment from **defined cardiovascular diseases** like hypertension, valvular disorders, and coronary artery disease
 - Etiology is **often** unknown **idiopathic**
- Common signs of heart failure include **shortness of breath**, unusual **fatigue**, and **swelling** in the legs, ankles, or feet caused by fluid buildup
 - Symptoms can develop slowly or start suddenly

Primary and secondary cardiomyopathies

- **Primary** cardiomyopathies involve mainly the heart
 - Can be **genetic** or **acquired**, such as acquired diseases like viral myocarditis, or anthracycline chemotherapy
 - Genetic causes include **mutations** in myocardial proteins of contraction, cell-cell contacts, or cytoskeleton (example, abnormal ion transport across cell membrane that can cause **dysrhythmias**)
- **Secondary** cardiomyopathies are a **component** of systemic or multi-organ disease processes, such as hemochromatosis or amyloidosis

Cardiomyopathy types

- **Dilated** Cardiomyopathy (DCM)
 - **90%** of cases
 - Abnormality = **Enlarged** chambers
 - Leads to **systolic contraction dysfunction**
 - Includes **Arrhythmogenic Cardiomyopathy**
- **Hypertrophic** Cardiomyopathy (HCM)
 - Abnormality = **Thickened** ventricular wall
 - Leads to **diastolic relaxation dysfunction**
- **Restrictive** Cardiomyopathy (RCM, next lecture)
 - **Least frequent**
 - Abnormality = **Stiff** myocardium
 - Leads to **diastolic relaxation dysfunction**



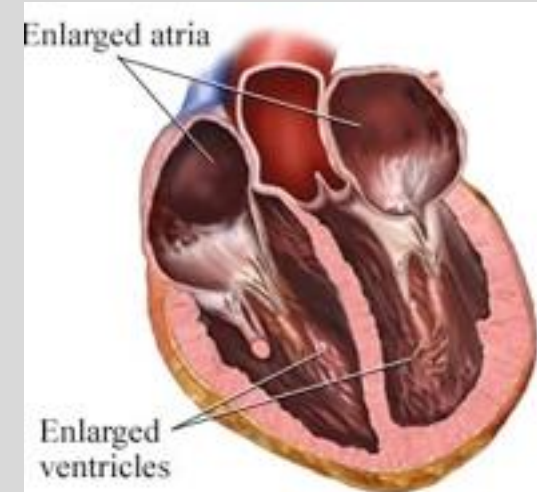
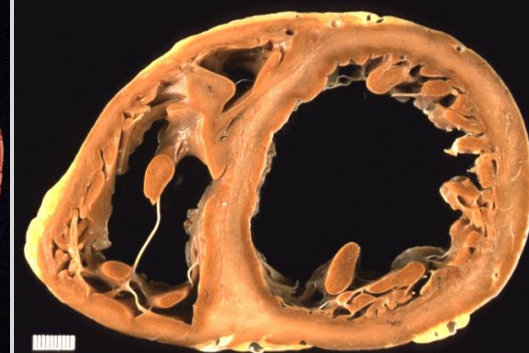
DCM

- **Eccentric hypertrophy** with progressive **dilation** and **contractile dysfunction** (ex. volume overload with new sarcomeres added in series)
 - In pressure overload like hypertension, added in parallel to existing sarcomeres (concentric hypertrophy)
- Usually, enlargement of all 4 chambers, **heavy, large, flabby** heart with marked dilation of all chambers, can be limited to just one side of the heart
- Commonly at diagnosis: End-stage disease
 - Suggested by left ventricular ejection fraction
 - **(LVEF) = <40%**, nl ~ 50-70%
- L heart failure effect R heart → **bi-ventricular** heart failure may result

DCM

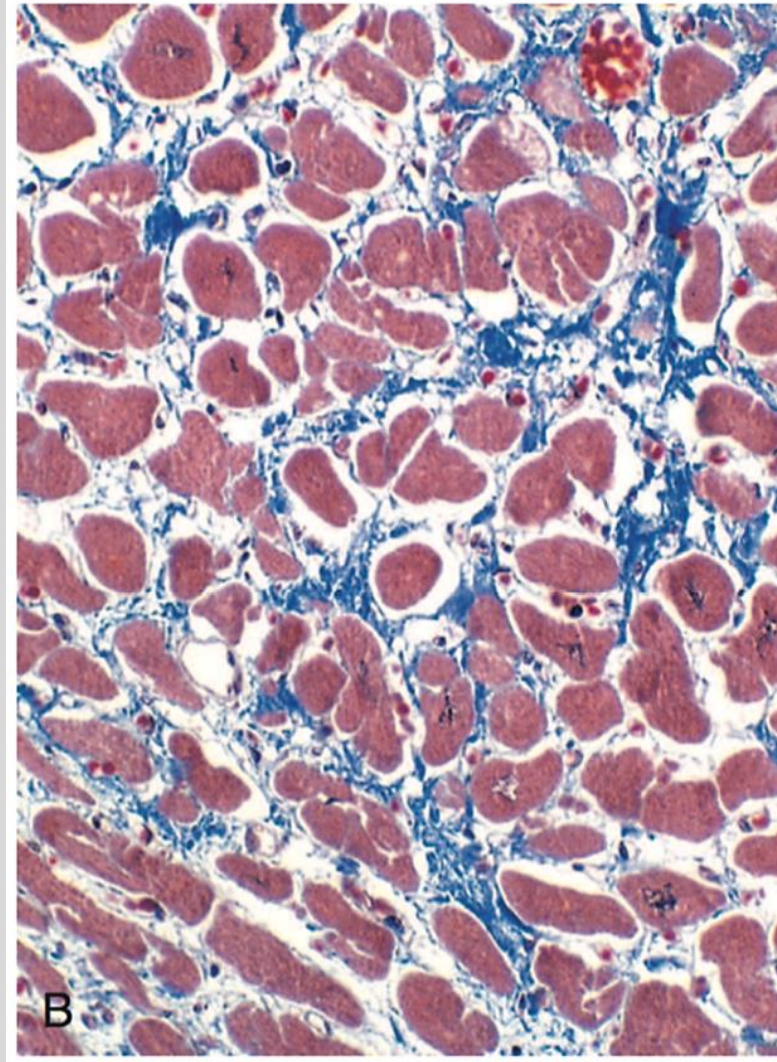


Normal



DCM has nonspecific histopathology

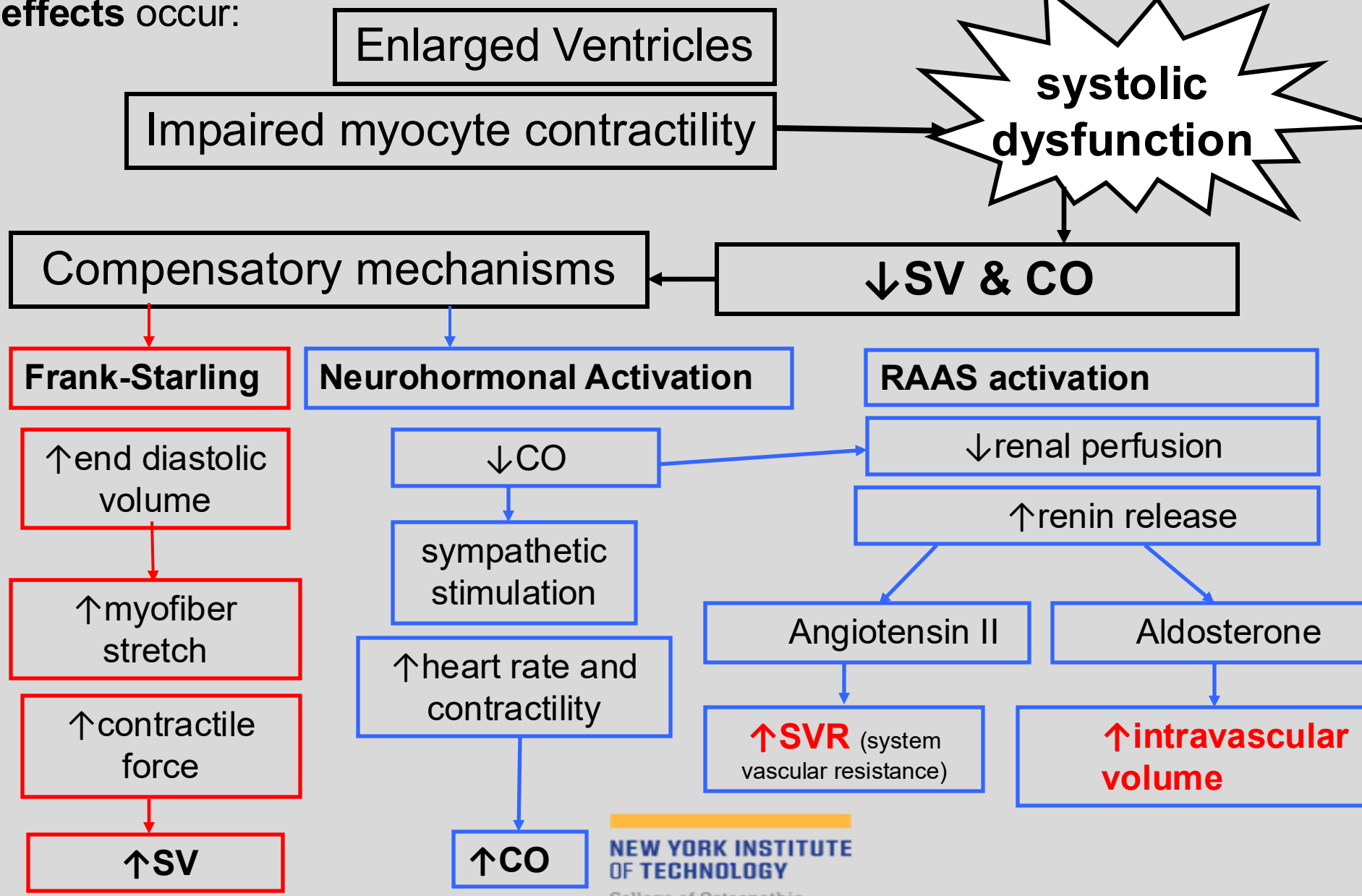
- Severity does **not correlate** with degree of dysfunction/prognosis, but may indicate etiology
 - **Myocytes** have **enlarged** nuclei & **irregular** hypertrophy
 - Other myocytes may be attenuated, stretched, irregular, degenerated, atrophic
 - Interstitial, endocardial, perivascular **fibrosis**



Trichrome stain - fibrosis appears blue

Robbins, Cotran & Kumar Pathologic Basis of Disease, 11th ed., 2026, Ch. 12

DCM Pathophysiology → ventricular **SV** (stroke volume) & **CO** (cardiac output) **decline** because of impaired myocyte contractility, three **compensatory effects** occur:



Compensatory mechanism effects

- Due to this, patient may be **asymptomatic early on**, but eventually **heart failure develops**
- **Angiotensin II** → ↑SVR → **afterload** becomes too high for **impaired LV**
- **Aldosterone** → ↑intravascular/EDV → fluid backs up → **pulmonary & systemic congestion**
- **Chronic** elevations in aldosterone and angiotensin II → cardiac **remodeling & fibrosis**
- With heart enlargement, **mitral/tricuspid regurgitation** may occur causing
 - Regurgitation of blood into the **LA** further **decreases** forward **stroke volume** during **systole** into aorta & systemic circulation
 - ↑**volume & pressure loads** in atria causing them to **dilate**; possible **atrial fibrillation**
 - Regurgitant volume returns to the **LV** during each **diastole** resulting in **greater volume load for LV** to pump out (dilation, LV **heart failure**)

DCM clinical features

- **Slow, progressive signs/symptoms of CHF**
 - **Signs** include cool extremities, **low arterial pressure**, **tachycardia**
 - Pulmonary venous congestion leads to auscultatory **crackles**
 - Left & upward displacement of **PMI**, **enlarged heart**
 - Third heart sound (**S3**) - blood rushing into a compliant, dilated, overloaded ventricle
 - **Normal** in **young** & athletes
 - **Abnormal** in **older** adults indicating **systolic heart failure**
- Complications: Mitral/Tricuspid **regurgitation**, atrial **fibrillation**, mural **thrombus/thromboembolism**
- Annual mortality high: 10-50% die usually from progressive cardiac failure or arrhythmia

DCM etiology general categories examples

- **Idiopathic (most common)**
- **Familial/Genetic**
- **Inflammatory**
 - Infectious (Viral)
 - Coxsackievirus A & B
 - Adenovirus
 - Chagas' disease
 - Noninfectious
 - Connective tissue diseases*
 - Peripartum cardiomyopathy*
 - Sarcoidosis*
 - Hemochromatosis
- **Toxic***
 - Chronic alcohol ingestion
 - Chronic cocaine use
 - Chemotherapeutic agents (eg. Doxorubicin)
- **Metabolic***
 - Hypothyroidism
 - Chronic hypocalcemia or hypophosphatemia
 - Beriberi
- **Neuromuscular**
 - Muscular or myotonic dystrophy

* Possibly reversible

DCM Pathogenesis

Peripartum associated:

- Occurs late in **pregnancy** or up to 6 months **postpartum** & may **recur**
 - Probably **multi-factorial** - **poorly understood** immune response, cytokine release, pregnancy associated hypertension, volume overload, nutritional deficiencies
- **50% regain normal** ventricular function
- **Risks** include advanced maternal **age**, African American **race**, **multiple gestations**

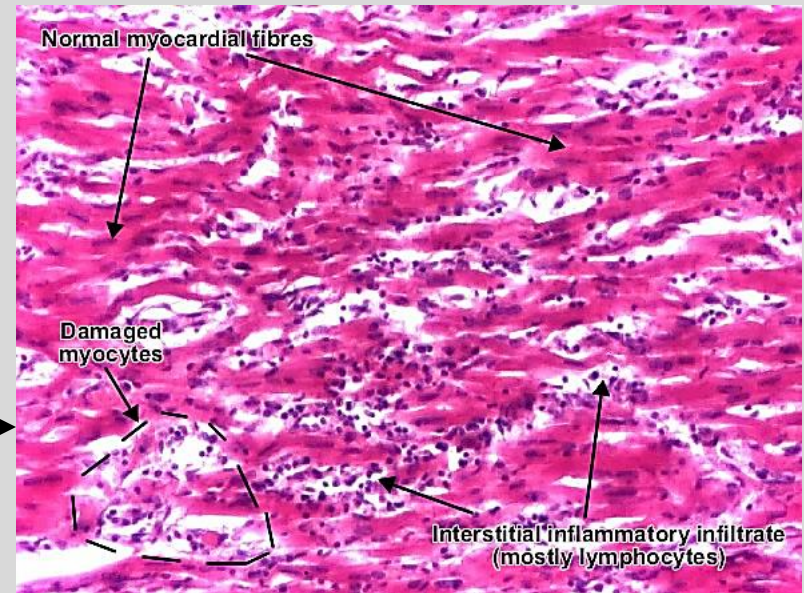
Genetic Causes:

- Familial in **30-50% of cases** - **AD** inheritance is most common
 - Mutation in *TTN* gene encoding **titan** may account for ~ 20% of cases
- Less commonly **X-linked**, **AR**, or **mitochondrial** inheritance
 - **X-linked DCM** linked to gene for **dystrophin** (links cytoskeleton to extracellular matrix) – **muscular dystrophy signs** after puberty to early **adult**
 - Deletions in **mitochondrial** genes → problems with FA oxidation or oxidative phosphorylation; typically **pediatric** manifestation

DCM: Pathogenesis

Myocarditis: Heart may grossly appear **normal or dilated**; in advanced stages, possibly mural **thrombi**

- **Viral** – most common
 - **Lymphocytic infiltrate**, injury by **direct cytopathic effect, inflammatory cytokines**, possibly **fibrosis**
 - Some fully **recover** or develop **DCM**, HF or death



<http://www.pathologyatlas.ro/acute-viral-myocarditis.php>

- **Nonviral causes** - *Trypanasoma cruzi* (**Chagas Disease**) - **10% die** during attack, others have **chronic** immune-mediated myocarditis (**trypanosomes**) → cardiac insufficiency 10-20 years later
 - *Trichinella spiralis* (**Trichinosis**) - most common **helminthic** myocarditis
 - *Toxoplasma gondii* (**Toxoplasmosis**), *Borrelia burgdorferi* (**Lyme Disease**), *Corynebacterium diphtheriae* (**Diphtheria**)
- **Noninfectious causes**
 - **Hypersensitivity** myocarditis - **perivascular** interstitial infiltrates: **lymphocytes**, macrophages, and a high proportion of **eosinophils**
 - **Giant cell** myocarditis - **widespread** inflammatory **infiltrate**: multinucleate **giant cells**, lymphocytes, eosinophils, plasma cells, and macrophages
 - Necrosis is present
 - **Poor prognosis**

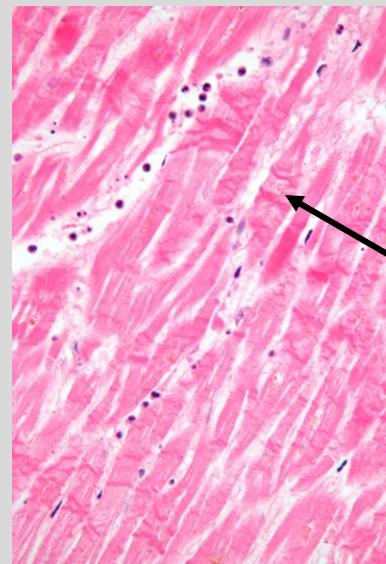
DCM pathogenesis

Toxin-Induced

- Strong association with **chronic excessive alcohol use**
 - Alcohol metabolite **acetaldehyde** = **direct toxin** to myocardium
 - **Ethanol (acetaldehyde)** → inhibition of mitochondrial oxidative phosphorylation and fatty acid oxidation → cell function declines
 - **Chronic** alcoholism → **thiamine** deficiency → **beriberi** - DCM & beriberi heart disease indistinguishable
 - Can be **reversible**: ventricular function improves with **cessation of alcohol**
- **Other toxins**: **Adriamycin** chemotherapy agents (**Doxorubicin**/Daunorubicin), cobalt, chronic **cocaine** use, **iron** overload - **hemochromatosis** (genetic or exogenous, ex. via transfusions)

Supraphysiologic Stress: Persistent **tachycardia**, **hyperthyroidism** lead to DCM

- Other conditions include **pheochromocytomas** (epinephrine), **cocaine** or vasopressor agents such as **dopamine** - **catecholamines** toxicity may cause multifocal myocardial **contraction band necrosis**
- Also, **intense autonomic stimulation** due to intracranial lesion or **emotional duress**
 - **Takotsubo** (“octopus pot”) **cardiomyopathy/ heartbreak syndrome** - left ventricular contractile dysfunction following extreme **psychological stress**
 - Left ventricular apex is most often affected leading to “**apical ballooning**”
 - Mechanism uncertain – maybe **calcium overload** or focal **vasoconstriction**



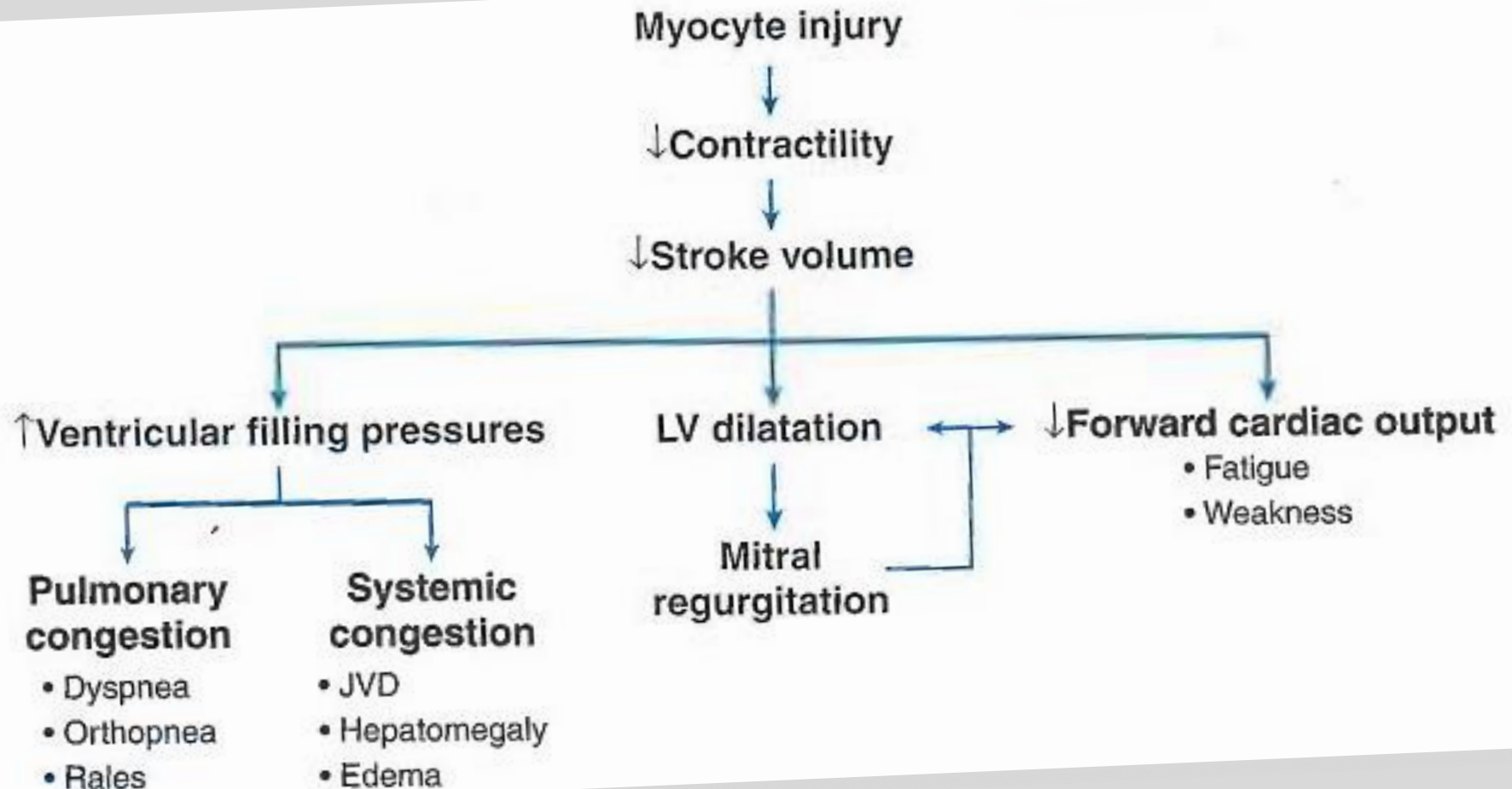
H&E stain: Contraction band necrosis

https://www.google.com/url?sa=t&source=web&rct=j&url=https%3A%2F%2Fen.wikipedia.org%2Fwiki%2FContraction_band_necrosis&ved=0CBkQjhxqFwoTCJj_5MaqnJYDFQAAAAAdAAAAABAY&opi=89978449

DCM treatment

- Goals: Treat **early**, treat **underlying cause**, **symptom** relief, decrease backward vascular **congestion**, increase forward **CO**, prevent complications, prevent **thromboembolic** events, prevent and treat ensuing **arrythmias**, improve long term survival
- Standard CHF **Rxs**: **Na⁺** restriction, **ACEI**/ARBs, **beta** blockers, **diuretics**, **digoxin**
- In those with poor LV function, anti- arrythmia drugs may worsen a weak heart pump and lead to arrythmias - **amiodarone** safest & does not carry a high risk of worsening heart failure
- **Anticoagulation** therapy may be for those prone to **thrombus** formation
- Implantable **cardioverter-defibrillator** (ICD) - prevents sudden cardiac death from dangerous heart rhythm
- **Cardiac transplant** may be the only definitive treatment
 - With no transplantation, 5 yr survival =74%; 10 yr = 55%
- 1/3rd of patients **spontaneously improve** after diagnosis

DCM: Summary



Lilly, L. (2011). Cardiomyopathies. In *Pathophysiology of heart disease: A collaborative project of medical students and faculty* (5th ed.). Baltimore, MD: Wolters Kluwer/Lippincott Williams & Wilkins.

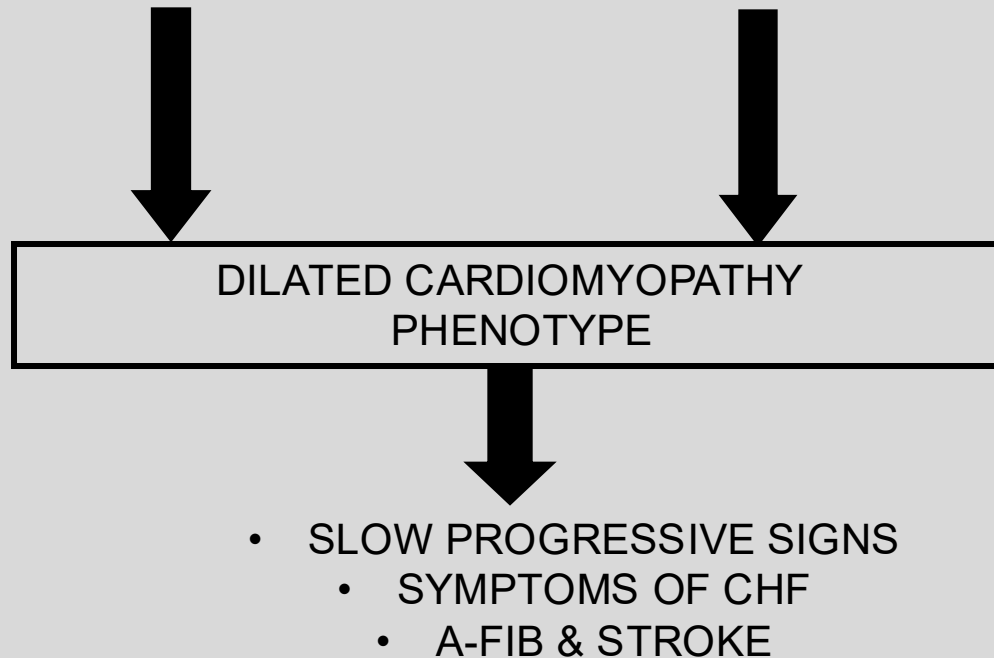
DCM: Summary

GENETIC (30 – 50%)

- DEFECT IN FORCE GENERATION
- FORCE TRANSMISSION
- OR MYOCYTE SIGNALING

NON-GENETIC

- MYOCARDITIS
- PERIPARTUM
- ALCOHOL
- ETC.

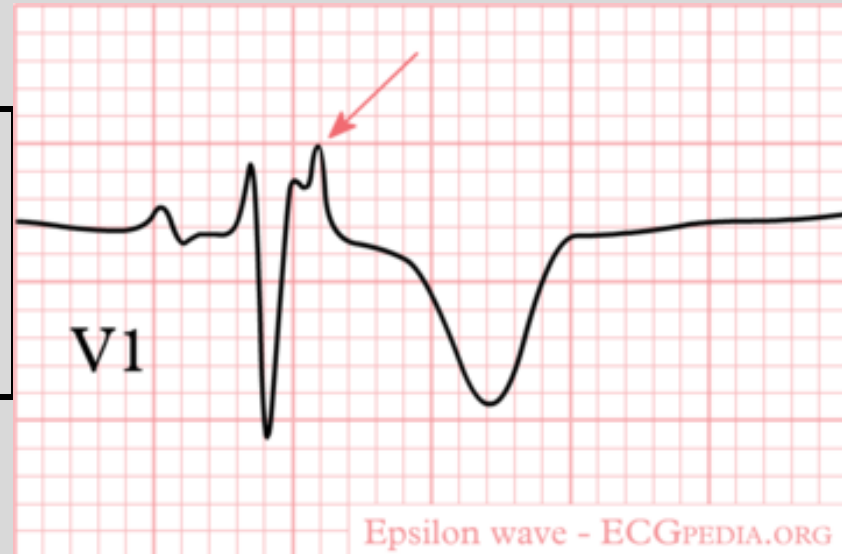


Arrhythmogenic Right Ventricular Cardiomyopathy (ARVC)

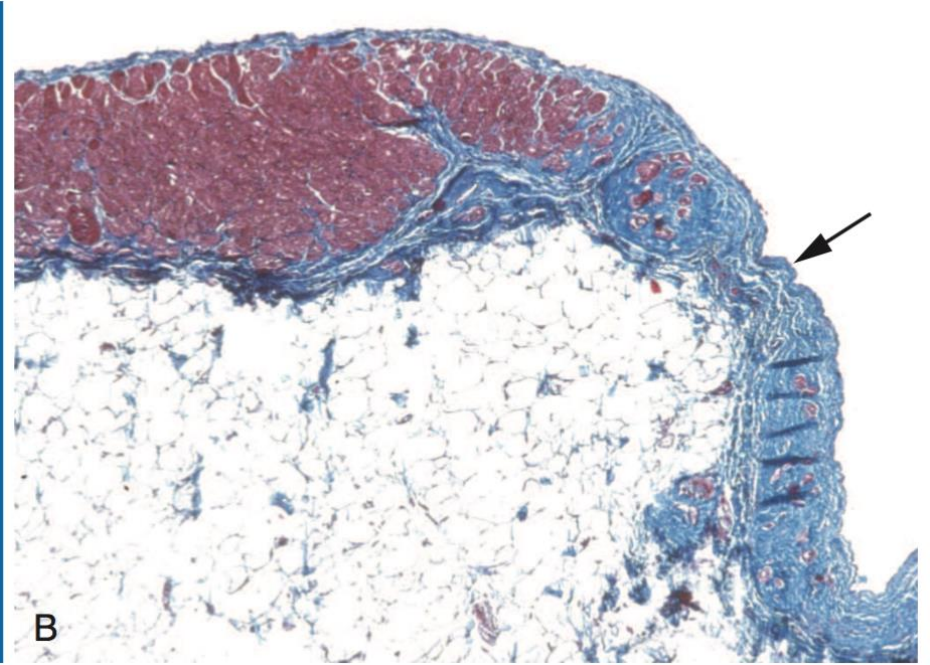
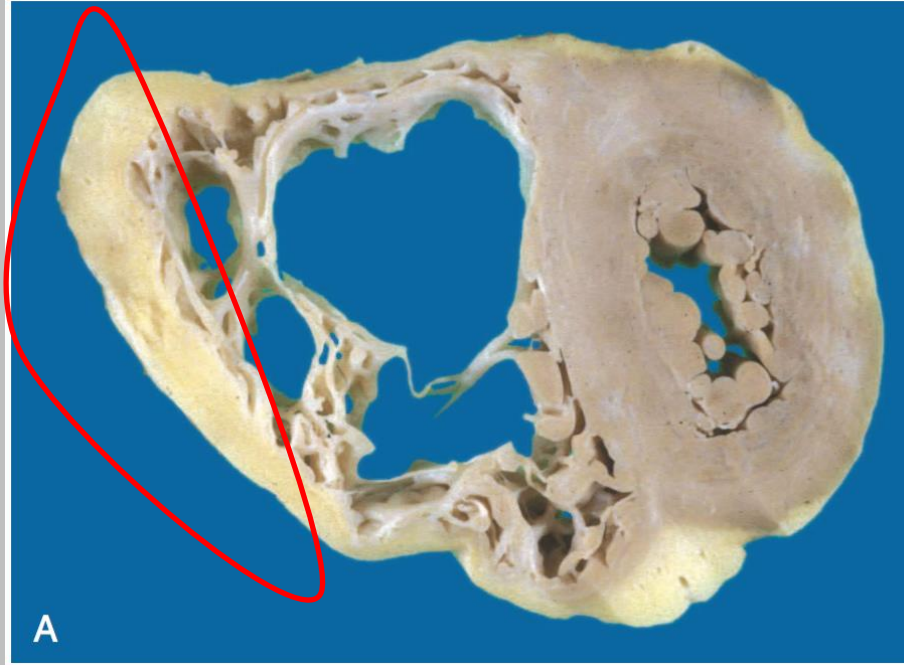
AKA Arrhythmogenic RV Dysplasia

- **Rare, progressive**, mostly sporadic
- Some familial forms occur: **AD** with variable penetrance - deletions of cardiac myocyte cell membrane **desmosome** junctional proteins → → **fibrosis** & myocyte loss w/ **fatty replacement**, or **aberrant signaling** between cells
 - Leads to **RV failure + rhythm disturbances**, esp. ventricular tachycardia and fibrillation with **sudden death** - associated with reentrant VT originating from the RV
 - Left sided involvement may occur
 - Can see **abnormal RV morphology** with non-invasive imaging
- Treatment usually involves **ICD** (implantable cardioverter defibrillator)
- **Naxos syndrome**: ARVC + **hyperkeratosis** of plantar/palmar surfaces associated with mutations in **plakoglobin** (desmosome-associated protein)

- Can pick this up on routine EKG
 - **Inverted T** waves in V1-V3
 - Possible **epsilon wave** = terminal QRS notch in V1
 - Indicates abnormal RV activation



ARVC pathology



- Trichrome stain (fibrosis blue): **Loss of myocytes** and myocyte replacement with massive **fatty infiltration + fibrosis** +/- inflammation of **RV** → **severe thinning** of RV wall

Hypertrophic Cardiomyopathy (HCM)

- Hypertrophic obstructive cardiomyopathy aka idiopathic hypertrophic subaortic stenosis **incidence** 1 in 500 births worldwide
- **Poorly compliant** left ventricular myocardium leading to **abnormal diastolic filling**
 - **Intermittent** ventricular outflow **obstruction** (~1/3rd) - **diastolic dysfunction**; **systolic** function is usually **preserved**
- Unexplained LVH in **absence of external stimuli** (e.g., HTN or aortic stenosis)
 - Disease of **sarcomere proteins**, causing massive **primary** myocardial hypertrophy of ventricular wall without dilation → ↑ muscle mass existing **vasculature cannot support** → ↓ compliance & outflow **obstruction**, coronary arteries also narrow due to wall hardening → **myocardial ischemia**
- Heart is thick-walled, heavy, and **hypercontracting vs** flabby, hypocontracting heart of **DCM**

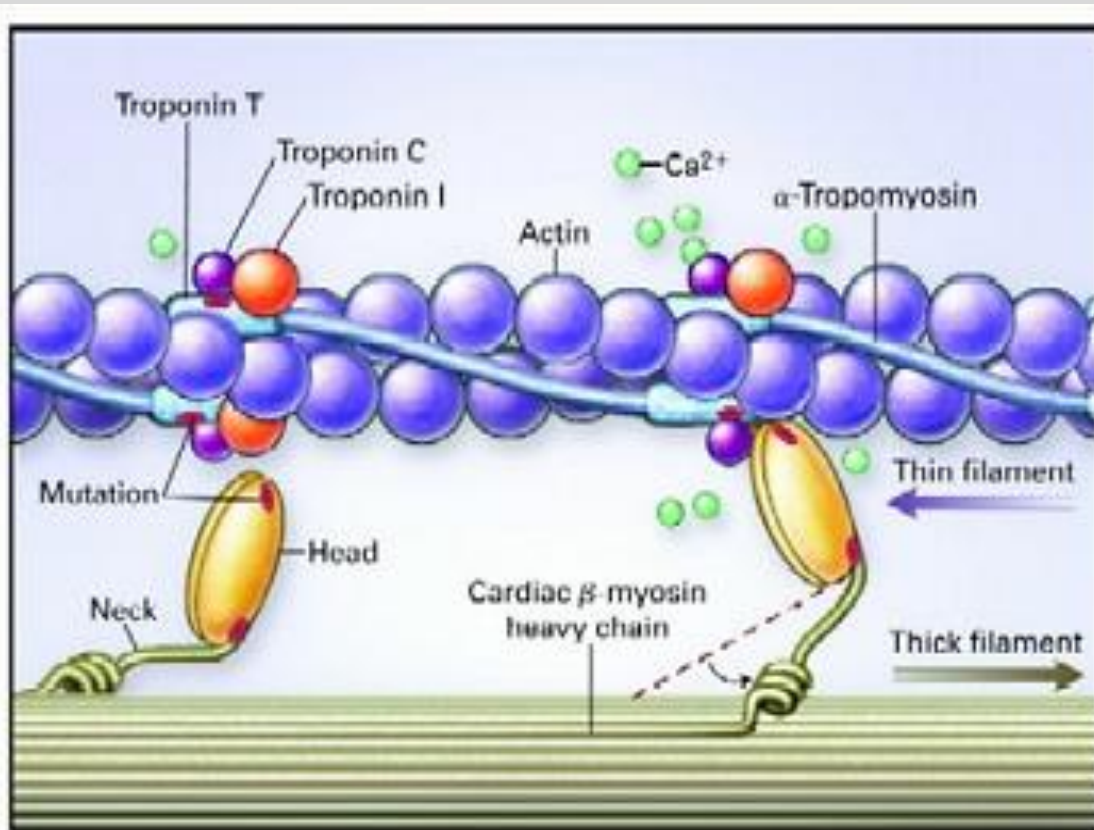
HCM etiology

- ~75% **familial** with **AD inheritance**; variable expression
- **100's** of known **mutations**, encoding for **sarcomere proteins**
 - Most are missense point mutations in 9 genes
 - **beta-myosin heavy chain (most common)**
 - cardiac troponins
 - myosin binding protein C
 - alpha-tropomyosin
 - associated w/ Friedreich Ataxia (often leads to severe heart complications)
- Sequence of events from mutations to disease poorly understood
- **Prognosis varies** widely, correlates with specific mutations

HCM pathogenesis

Normal Sarcomere Contraction

- Ca^{2+} binds to **troponin** C, I, and T and α -tropomyosin → **myosin-actin** interaction
- Actin stimulates **ATPase** activity in myosin head → force generation



Disease

Mutated protein within sarcomere

contractile impairments

increased myocyte stress

Compensatory
Hypertrophy

Growth factor
release

Fibroblast
proliferation

Myofiber disarray

Interstitial fibrosis

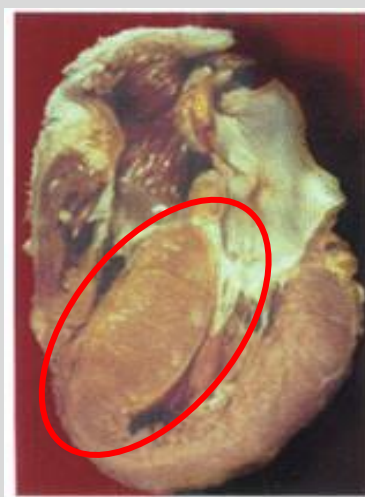
Illustration of the sarcomere
(Kamisago, M. et al. N Engl J Med.
2000; 343:1688-96. Copyright 2000
Massachusetts Medical Society. (All
rights reserved.)

HCM pathology & histopathology

Normal LV

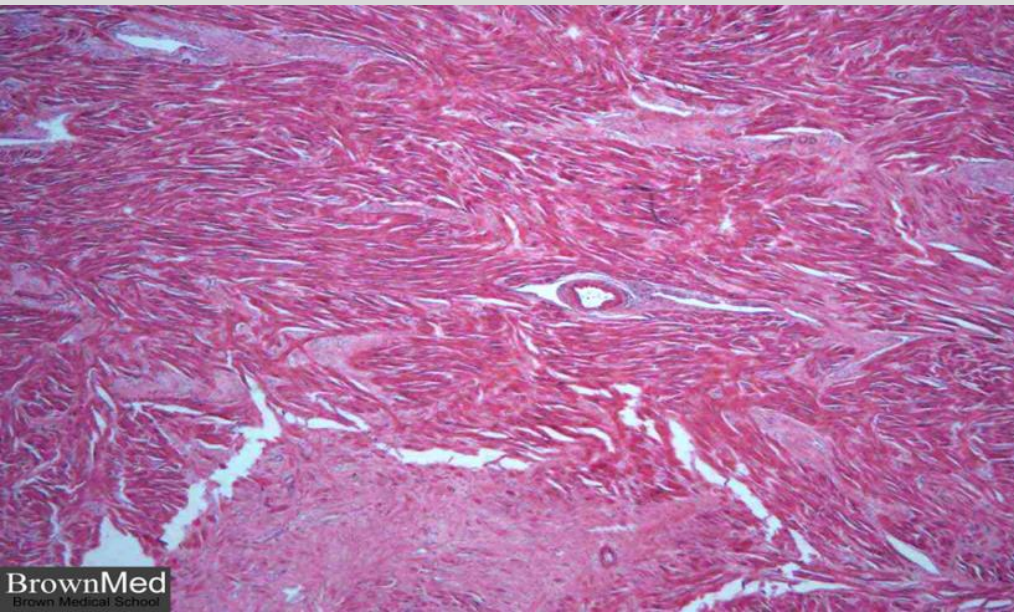


HCM



- **Hypertrophy** is usually not concentric: **Septum > free wall** thickness ($\geq 3:1$) - asymmetric & septum more common
 - Subaortic septum most commonly affected
 - LV lumen has a “**banana like**” shape due to protrusion of septum
 - The **LVOT** (outflow tract) often has a fibrous **endocardial plaque** due to thickening of anterior **mitral leaflet**
- Chaotic arrangement of myocardial **fibers** makes the tissue **stiff** and leads to **diastolic failure**

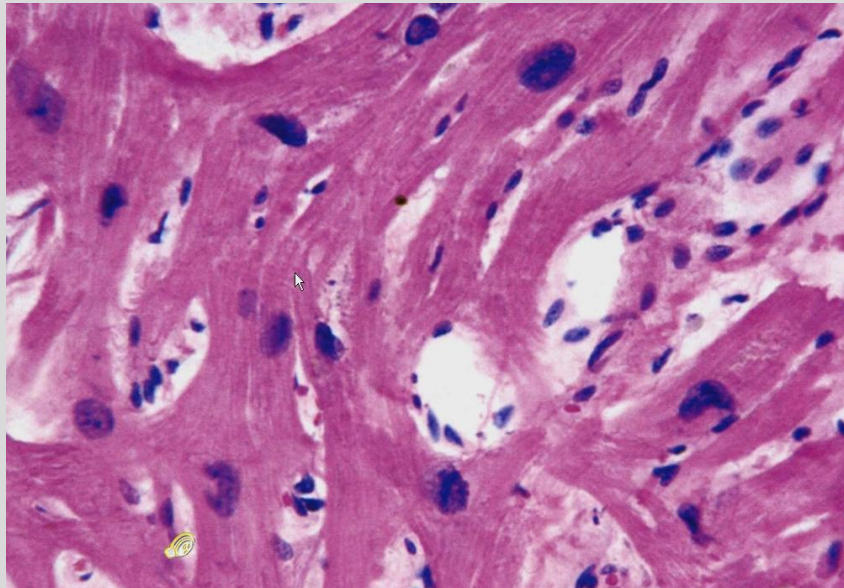
Source: The failing heart. J. A. Towbin and N. E. Bowles, Nature 415, 227-233(10 January 2002)



- Septum: A case of HCM
- **Myofiber disarray** (bundles running in all directions)
- Interstitial **fibrosis** is also present

HCM histopathology

- Hallmarks of histological diagnosis include **myocyte hypertrophy**, **fibrosis** and **disarray**
 - **Haphazard myofibril disarray** of bundles of extremely hypertrophied myocytes, and contractile elements in sarcomeres within myocytes
- Interstitial and replacement **fibrosis**
- Aberrant myofibers present **in conduction system** may lead to fatal **arrhythmias**



HCM pathophysiology

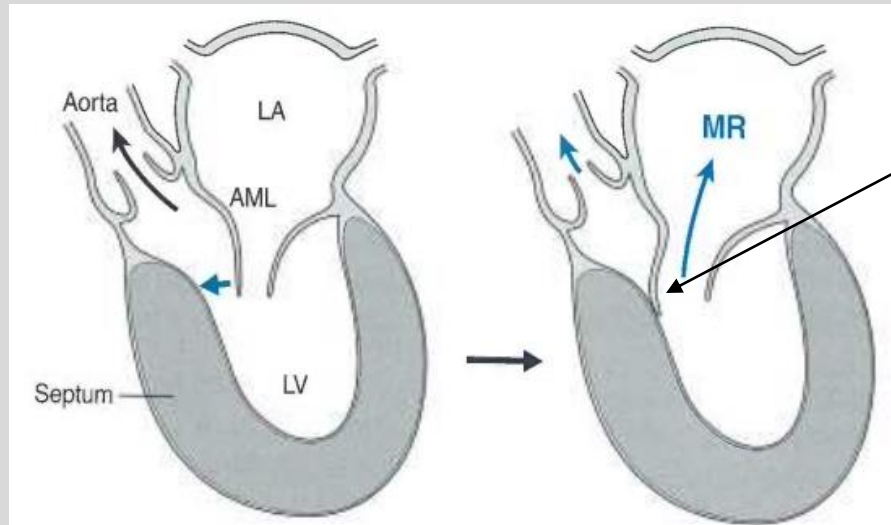
Without Obstruction

- ↓ LV chamber size
- ↓ LV compliance: SV & CO ↓
- **Diastolic pressures ↑** & transmits backwards to **LA** & **pulmonary** vessels

With Obstruction

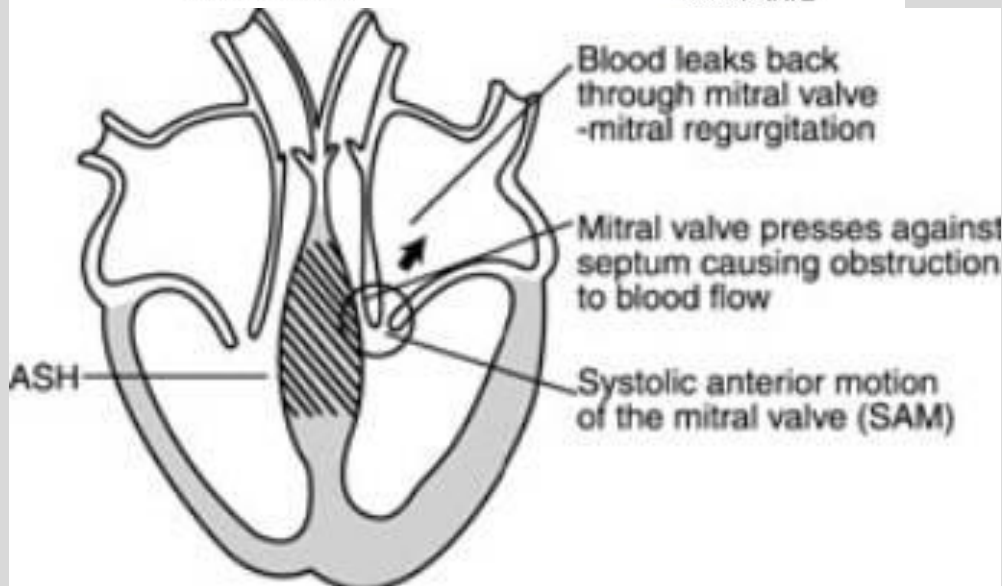
- = **Dynamic** obstruction
- Systolic contraction: Thickened septum narrows LV outflow track → more rapid ejection of blood creates a force that draws the anterior leaflet of mitral valve towards septum → **transient obstruction** of **LVOT**

HCM pathophysiology



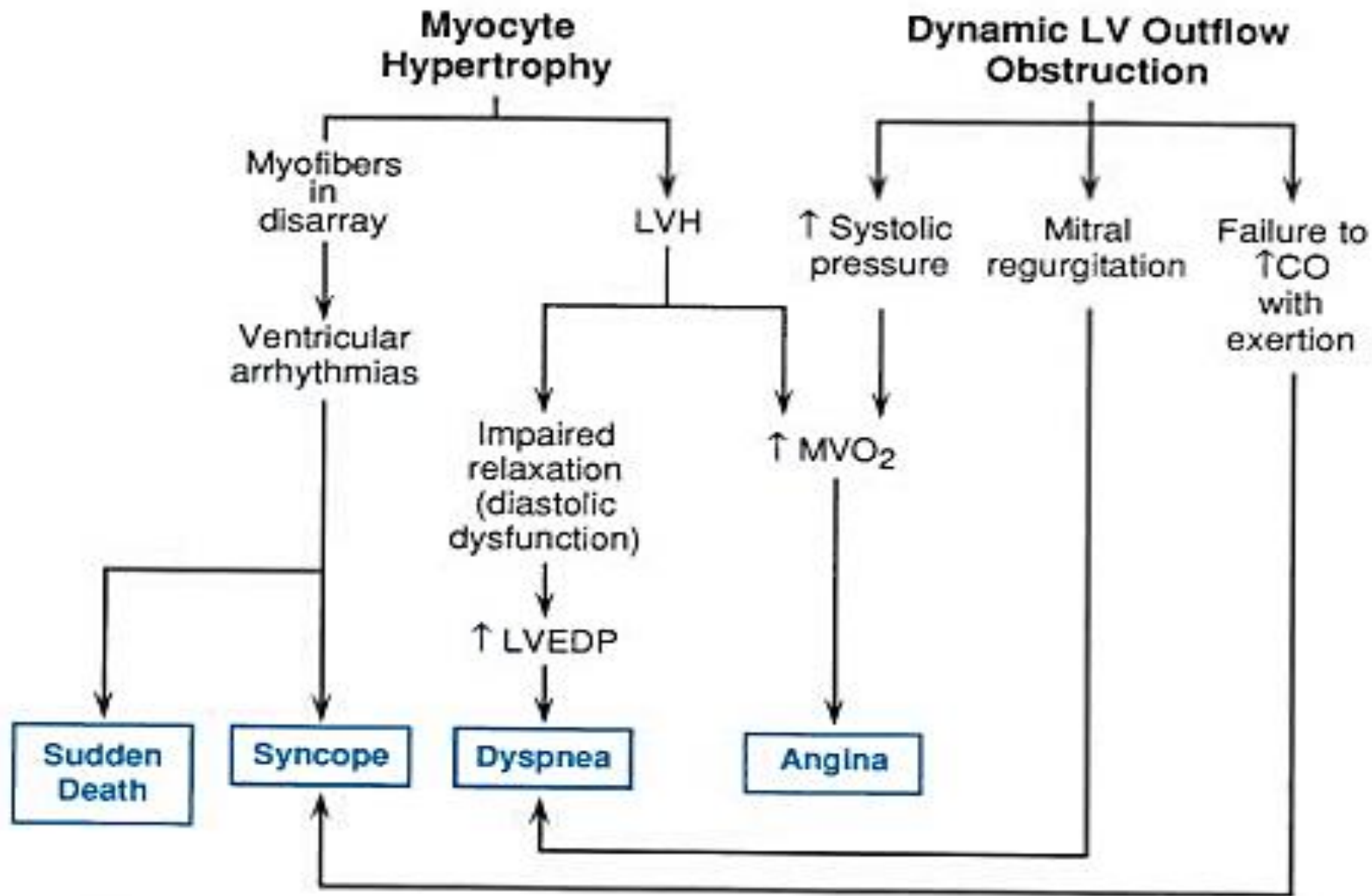
- **Septal subaortic hypertrophy** → **functional aortic stenosis** below the aortic valve → **Mitral Regurgitation** & atrial **fibrillation**

Both have similar results: **Pressure gradient** ↑ wall stress & myocardial O₂ consumption → angina & focal **ischemia**, dyspnea, possible syncope



- **Hypertrophic obstructive cardiomyopathy**: Asymmetric septal hypertrophy (**ASH**), systolic anterior motion (**SAM**) of the mitral valve leaflet & obstruction of LVOT
- Maybe mitral regurgitation also

HCM pathophysiology summary



HCM clinical features

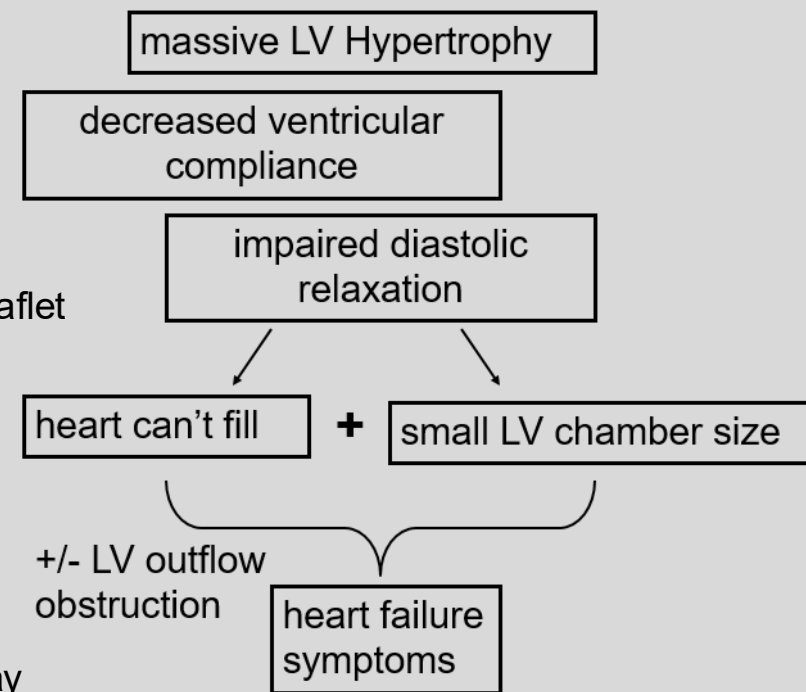
- From no to extensive limitations, **physical activity varies**
- Most do **not** have severe **outflow obstruction**
 - Most **symptoms** result from hypertrophy-induced **impairment in diastolic filling**
- Major clinical problems/complications: Mitral **regurgitation**, atrial **fibrillation** w/ mural **thrombus**, cardiac failure, ventricular arrhythmias, **syncope & sudden death**
 - **Most common** cardiac abnormality in **young athletes** who **die suddenly** during exertion (**ventricular arrhythmia**)
 - **Risks** = History of **syncope**, **family history** of sudden death, high risk **mutations**, extreme hypertrophy (**LV wall thickness** correlates with risk)

HCM: Clinical features mechanisms

	Hypertrophy	Obstruction
	Mechanism	
Dyspnea	diastolic impairment → ↑ LV, LA, PCW pressures	↑ systolic LV pressures during obstruction + possible MR
Angina	hypertrophy → ↑ muscle mass → ↑ O ₂ demand hypertrophy → squeezes down on small coronary branches in ventricular wall → impaired myocardial perfusion	Plus ↑ LV pressure during systole → ↑ wall stress → ↑ myocardial O ₂ demand
Light Headedness		standing → blood pools in legs → ↓ preload → ↓ LV return → ↓ LV chamber size → ↑ obstruction → transient ↓ in CO and cerebral perfusion
Syncope/ Arrhythmias Arrhythmias can worsen HCM symptoms like a. fib. causing loss of atrial kick so diastolic filling is further impaired and pulmonary congestion is worsened	Aberrant myocytes within conduction system → abnormal conduction & development of arrhythmias → syncope	↑ physical exertion → ↑ force of contraction worsens obstruction → transient outflow block → CO ↓ → syncope

HCM pathophysiology summary diagram - Rxs

- **Things that ↓ LV cavity** volume will bring anterior mitral leaflet & septum closer together
 - ↓ **venous return**
 - **Blood volume ↓**
 - **Positive inotropic drugs** ↑ force of contraction
 - So, this **promotes** obstruction
 - → Meds, surgery, ICD, avoid very heavy **exercise**, stay **hydrated**
- **Things that ↑ LV cavity** volume will push anterior mitral leaflet & septum apart
 - ↑ **venous return**
 - **Blood volume ↑**
 - **Negative inotropic drugs** ↓ force of contraction
 - So, this **reduces** obstruction
 - → Meds, surgery, ICD (for dysrhythmia), avoid very heavy **exercise**, stay **hydrated**



HCM clinical exam & treatments

Physical Exam

- **S4 heart sound** (LA contraction into **stiffened LV**)
- Outflow obstruction: **Harsh systolic ejection murmur**, crescendo, de-crescendo, caused by turbulent flow through the narrowed outflow tract; best heard at **left lower sternal border**

Screening

- **First degree relatives**
- **Serial echocardiograms** to note increases in hypertrophy during growth
- Should evaluate all HCM patients, even if asymptomatic - at risk for **sudden death**

Treatment

- Most do well with medications that relax the heart:
 - Standard therapy = **beta blockers**
 - Non-dihydropyridine Ca^{2+} **channel blockers** (verapamil)
 - Antiarrhythmic drugs such as **amiodarone**
- Avoid strenuous **exercise & drugs** that ↓ **preload** or ↑ **force of contraction**
- Implantable Cardioverter - Defibrillator (**ICD**): For patients at risk of sudden cardiac death(from dysrhythmias)
- Surgery: **Myomectomy** (more proven to hold) or septal **ablation** with alcohol

HCM summary

- Most commonly genetic
- **SARCOMERIC PROTEIN MUTATION**
- DEFECT IN ENERGY TRANSFER
- OR MITOCHONDRIA
- AND/OR SARCOMERIC DYSFUNCTION
- **Versus** defect in force generation and/or transmission in **DCM**



HYPERTROPHIC CARDIOMYOPATHY PHENOTYPE



- HEART FAILURE
- A-FIB & STROKE
- SUDDEN DEATH

Thank you,
Lecture Feedback Form

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>